

28 – M8 Reporting Pipeline

Module design for the pipeline, not the reports: projections built from the event stream into a reporting schema on the replica, end-of-day aggregates keyed to each shop's day-close, a per-entity freshness stamp on every federation-level figure, a decision-driven report catalogue rendered through the A4 pipeline, spreadsheet export, the analyst SQL role, and the exception queues. Individual report definitions arrive from the pending workshop and slot into this pipeline without changing it.

Field	Value
Document	28 – M8 Reporting Pipeline (Stage 2 module design; template of doc 20 §5)
Version	v0.1 – draft for design review; contract freeze targeted end of Stage 2 week 6
Date	9 September 2026
Owning developer	Module developer (full-stack); the report-catalogue workshop supplies definitions as data
Inputs	05 SRS (FR-REP-010–050); 07 §5 placeholder report set; 16 answers (O section, F-09, M-04, M-06, N-06); 17 v0.2.1 §7.8, ADR-30; 18 v0.2 part E; 19 v0.2 §4.3, §6, §9, DR-3; 20 v0.3 M8 and Appendix A (analyst SQL role); 21–27 event contracts; 30 §5.8
Depends on	Every module's events (through the outbox archive), M1 scope and RLS classes, kernel (event backbone, A4 renderer, scheduling, audit exception queues)
Depended on by	M9 (exports to notifications), doc 31 fleet console (health projections), M4 (delivery performance), M10 (phase-2 forecasting inputs)

Three commitments. Every projection is rebuildable from the outbox archive, so a wrong report is fixed by a rebuild, never by editing figures. Every federation-level number carries the freshness of its least-recent contributor, because some shops are always offline and a figure without that stamp misleads. And the catalogue is decision-driven: a report enters it with the decision it supports and the audience that makes it (O-01), or it does not enter at all.

1. Purpose and scope

In scope (phase 1)	Out of scope
Projections: fact and position tables in a reporting schema, maintained by idempotent event consumers, rebuildable from the archive (ADR-30)	Modules reading projections for transactional decisions (they never do)
Freshness: per location and per entity, stamped on every figure and every rendered report (O-03, FR-REP-020)	An anonymised aggregate tier (M-06: not needed)
End-of-day aggregates keyed to each location's day-close; monthly roll-ups; comparability across years (O-07)	–
Report catalogue as data: definition, parameters, scope class, decision supported, audience, output formats; the doc 07 §5 placeholder set until the workshop finalises (O-01)	Self-service report builder (backlog, O-05); mobile dashboard app (O-06)
Rendering and export: A4 through the kernel pipeline in the reader's language; spreadsheet export as a job with a download link (O-05)	–
Dashboards: fixed tiles per scope with freshness; drill to catalogue reports	Real-time streaming dashboards beyond near-real-time projections
Exception queues: the REVIEW/ALERT audit events by scope with acknowledgement, as the reporting face of doc 19 §4.3 (N-06)	The audit store itself (kernel)
Analyst SQL role on the replica under the Federation view (doc 20 Appendix A)	Write access of any kind; direct access to operational schemas
Alias resolution (M2) so promoted SKUs keep unbroken trends (B-10)	–
Phase 2 designed as projections: stock ageing, reorder suggestions, ABC classes, demand inputs for M10	A columnar store in phase 1 (trigger-based, doc 19 DR-3)

2. Requirements coverage

Identifier	Requirement (abridged)	Design section
FR-REP-010	Near-real-time operational views from the event stream; heavy aggregates end-of-day (O-03)	3.1, 3.3, 9.1
FR-REP-020	Per-entity data-freshness stamp on every federation-level figure	3.2
FR-REP-030	Offline shop reports on the till (Z-report, day sales, session)	M6 (doc 26 §3.5); referenced only
FR-REP-040	Decision-driven catalogue; fixed set; finalised in the workshop (O-01)	3.4, 6.1
FR-REP-050	Spreadsheet export; printed A4 reports (O-04, O-05)	3.5
O-07	Indefinite history; comparable trends across years	3.3, 9.2
F-09	Shrinkage percentage per shop visible to the Federation	3.1 (loss projections), 3.6
M-04	Federation sees margins, profitability and shrinkage view-only	3.6
N-06	Exception reports with named reviewers	3.7
B-10	Alias resolution preserves history	3.1
ADR-30	Outbox archive as the replay source	3.1, 4.2
Doc 20 Appendix A	Analyst SQL role	3.8
Doc 19 DR-3	Columnar store trigger	9.3

3. Domain model

Schema reporting on the primary (written by consumers) and read on the replica. Nothing here is a source of truth; every table can be dropped and rebuilt.

3.1 Projections

Projection	Grain	Built from	Serves
receipt_fact / receipt_line_fact	One row per receipt and per line: location, business date, operator, SKU (alias-resolved), batch, qty, prices, ceilings applied, rule, tax, tender kinds	receipt.issued/voided/refunded.v1	Sales reports, markdown cost, cap analysis, VAT summary
sales_daily	Location × business date × SKU (and roll-ups by tag, entity, district)	receipt facts at day-close	Daily sales, trends, ranking
stock_position	Location × SKU × batch × condition: on hand, cost, expiry, days held	stock.moved.v1 (M5)	Stock position, ageing (P2), expiry
stock_movement_fact	One row per movement with type and cost	stock.moved.v1	Stock cards, cost of goods sold, movement analysis
loss_fact	Write-off lines by category, value, location, approver	writeoff.posted.v1, count adjustments, negative-lot clearances	Loss by category, shrinkage %, tolerance analysis
trade_document_fact	Orders, DNSs, GRNs, invoices, credit notes, receipts between entities with status and amounts	M4 events	Fill rate, delivery performance, receivables, statements
dispute_register	Discrepancies and claims with windows, resolution, arbitration	M4 events	Dispute register, escalation ageing
exposure_snapshot	Relationship × day: exposure components	exposure.updated.v1	Exposure trends, limit utilisation
customer_receivables	Account × day: balance and ageing buckets	M7 events	Debtors, ageing
till_session_fact	Sessions with float, tenders, drops, variance	M6 session events	Till variance, cash control
pricing_review_fact	Reviews raised by M3 with resolution	pricing.review_raised.v1	Pricing hygiene
fleet_health	Device × heartbeat: version, snapshot age, pending, peripherals	till.heartbeat.v1, device.*	Fleet console (doc 31), sync health
location_freshness	Location: last applied device seq time; entity roll-up	sync.batch_received.v1 (kernel)	Every stamp

Aspect	Design
Idempotency	Each projection consumer upserts by the event's aggregate id and event id (kernel inbox); a replayed event produces the same rows.
Alias resolution	Facts store sku_id as issued and canonical_sku_id resolved through M2's alias at projection time; reports group by the canonical id so promotion and merge never break a trend (B-10).
Position tables	Recomputed incrementally from movements; a nightly comparison against M5's lots raises a REVIEW on drift (the same discipline as doc 25 §3.2).

3.2 Freshness

Aspect	Design
Per location	location_freshness.applied_through = the received_at of the last event applied from that location's devices; updated by the kernel's sync.batch_received.v1.
Per entity and federation	The minimum over the contributing locations; a figure covering N locations carries (freshness = min applied_through, locations_stale = count older than threshold).
On every output	Dashboards, catalogue reports and exports embed the stamp in the rendered output and in the API response; the FreshnessStamp component (doc 30) renders it as 'includes shops synced up to hh:mm; 3 shops not yet synced'.

3.3 End-of-day and roll-ups

Aspect	Design
Trigger	<code>location.day_closed.v1</code> (kernel day-close, doc 19 §9) per location; not a clock time for all shops. Late-arriving events for a closed day (a till that syncs two days later) update the day's aggregates and re-stamp freshness; the day is never 'final' before its freshness covers all its locations.
Aggregates	<code>sales_daily</code> , <code>cash_daily</code> , <code>loss_daily</code> ; monthly roll-ups by a nightly job; financial-year roll-ups by entity settings (M1).
Comparability	Facts keep the issued-time attributes (price, tax rate, MRP, rule) so a 2028 report on 2026 data reproduces 2026's figures; alias resolution keeps SKU continuity (O-07).

3.4 Report catalogue

Aspect	Design
Definition as data	<code>report_definition(report_id, name (three languages), decision_supported, audience_roles, scope_class (OWN FEDERATION_VIEW PARTY), parameters schema, data_source (projection query or SQL template), a4_template, xlsx_layout, freshness_rule, status, version)</code> . Registered by the Federation; entities cannot author definitions in phase 1 (O-05).
Phase-1 set	The doc 07 §5 placeholders, each with its decision and audience: daily sales by shop (shop manager: what sold, cash to bank); sales by SKU and tag (MPCS: what to reorder, what to drop); till sessions and variance (supervisor: cash control); stock position and expiry (shop and MPCS: what to count, mark down or move); loss by category and shrinkage % (MPCS and Federation: where losses arise); discrepancy and claim register (both parties: what is unresolved); discount and markdown cost by rule (MPCS: what promotions cost); VAT summary (accounts: what to file); exposure and statements (seller accounts: who to chase); receivables ageing (MPCS: khata risk); order fill rate and delivery performance (Federation and distributors: service levels); fleet and sync health (DevOps and Federation: which shops are dark). The workshop replaces or extends this list as data.
Runs	<code>report_run(run_id, report_id, requested_by, scope, parameters, freshness, status, object_key, rendered_at)</code> ; every run is audited (a report is a read of scoped data).

3.5 Rendering and export

Aspect	Design
A4	Definitions bind to templates rendered by the kernel A4 pipeline (doc 19 §5.3) in the reader's language with English fallback; output to object storage; <code>report.rendered.v1</code> .
Spreadsheet	<code>export_job(job_id, report_id or projection, scope, parameters, status, object_key, row_count)</code> produced by a worker with the caller's scope applied at query time; <code>export.ready.v1</code> ; downloads expire after 7 days; large exports paginate into multiple sheets.

3.6 Visibility on projections

Aspect	Design
Classes	Projection tables carry <code>owner_entity_id</code> and <code>location_id</code> and are under the same four RLS classes (doc 18 §3.7): a shop-scoped user sees only its location; an MPCS its locations; the Federation everything view-only including margins and shrinkage (M-04); PARTY sees trade documents it is party to through masked columns (no cost). Cost and margin columns are absent from any row a PARTY class can read.

3.7 Exception queues

Aspect	Design
Face of the audit store	<code>exception_item</code> is a projection of REVIEW/ALERT audit events with <code>reviewer_role_template</code> , acknowledgement state (from AUDIT_REVIEWED events) and escalation age; queues per entity and per named reviewer; acknowledgement is a kernel command (doc 19 §4.3), not an M8 write.

3.8 Analyst SQL role

Aspect	Design
Access	A database role <code>app_analyst</code> on the replica with SELECT on <code>reporting.*</code> only, connections through the pooler with the FEDERATION_VIEW session scope fixed by a login hook; issued to named Federation analysts holding <code>rpt.analyst.query</code> ; every session start is audited; no operational schema is grantable.

4. State machines

4.1 Report run

From	To	Trigger / command	Guard	Permission	Audit
–	REQUESTED	RequestReport(report, scope, parameters)	definition ACTIVE; caller's scope class permitted by the definition; parameters valid	rpt.report.run (per definition audience)	REPORT_REQUESTED
REQUESTED	RENDERING	worker picks up	–	system	–
RENDERING	READY	render complete	object stored; freshness embedded	system	REPORT_RENDERED
RENDERING	FAILED	error	reason	system	REPORT_FAILED (ALERT after 3)

4.2 Export job

From	To	Trigger / command	Guard	Permission	Audit
–	QUEUED	RequestExport(source, scope, parameters)	as report run; row estimate under the cap or paginated	rpt.export.run	EXPORT_REQUESTED
QUEUED	READY / FAILED	worker	–	system	EXPORT_READY / FAILED
READY	EXPIRED	Clock (7 days)	object deleted	system	–

4.3 Projection

From	To	Trigger / command	Guard	Permission	Audit
LIVE	REBUILDING	RebuildProjection(name, from_seq)	DevOps; replays from the archive into a shadow table	sys.projection.rebuild (MFA)	PROJECTION_REBUILD_STARTED
REBUILDING	LIVE	swap shadow into place; consumer resumes from the live position	row counts reconciled	system	PROJECTION_REBUILT
LIVE	LAGGING	consumer lag over threshold	–	system	PROJECTION_LAGGING (ALERT)

4.4 Report definition

From	To	Trigger / command	Guard	Permission	Audit
–	DRAFT	RegisterReportDefinition	caller Federation; decision and audience stated; data source validates against the schema	rpt.definition.manage (F)	REPORT_DEFINED
DRAFT	ACTIVE	ActivateReportDefinition	template renders on fixture data; scope class set	rpt.definition.manage (F, MFA)	REPORT_ACTIVATED
ACTIVE	RETIRED	RetireReportDefinition	reason	rpt.definition.manage (F)	REPORT_RETIRED

5. Contract

5.1 Commands

Command	Input (essentials)	Permission	Events
RegisterReportDefinition / ActivateReportDefinition / RetireReportDefinition	definition (names, decision, audience, scope class, parameters, source, templates)	rpt.definition.manage (F)	report_definition.*.v1
RequestReport	report id, scope, parameters, language	per definition audience	report.requested/rendered/failed.v1
RequestExport	report or projection, scope, parameters	rpt.export.run	export.ready.v1

Command	Input (essentials)	Permission	Events
RunEndOfDay (internal)	location, business date	system (from day_closed)	end_of_day.completed.v1
RebuildProjection	projection, from_seq	sys.projection.rebuild (MFA)	projection.rebuilt.v1
PinDashboardTile (per user)	tile id, position	any	–

5.2 Queries

Query	Returns	Scope rule
GetDashboard(scope)	Tiles with figures and freshness	Caller's scope class
GetReportData(report, scope, parameters)	Tabular data with freshness (the same source the renderer uses)	Definition's scope class
ListReportDefinitions(audience)	Catalogue visible to the caller	Any
GetReportRun / ListRuns(scope)	Run status and download link	Requester; OWN
GetFreshness(entity location)	applied_through, stale locations	Any authenticated
GetExceptionQueue(scope, severity, since)	Items with acknowledgement state	Named reviewer roles; OWN
ProjectionStatus()	Lag, last seq, rebuild state per projection	DevOps; FEDERATION_VIEW
DeliveryPerformance / FillRate(seller, period); ShrinkagePct(location, period); Receivables(mpcs)	Named projections used by other modules' screens	OWN, FEDERATION_VIEW

5.3 Events published

Event	Payload	Consumers
report.requested/rendered/failed.v1	run, report, scope, object key, freshness	Notifications (M9), audit
export.ready.v1	job, object key, rows	Notifications
end_of_day.completed.v1	location, business date, totals, freshness	M7 (banking expected), M9, dashboards
projection.rebuilt.v1 / projection.lagging.v1	projection, positions	DevOps alerts
exception_report.generated.v1	scope, counts by severity, reviewer	Notifications to named reviewers (N-06)

6. Key flows

Flow	Sequence	Failure paths
6.1 Adding a report after the workshop	Federation registers a definition as data: names, the decision it supports, audience roles, scope class, parameter schema, a projection query, an A4 template and a spreadsheet layout → validated against fixture data → activated (MFA) → appears in the catalogue for its audience	A definition without a decision or audience is refused; a query touching a non-reporting schema is refused
6.2 A shop manager's morning	Dashboard tile 'yesterday' shows sales, cash to bank, variance with freshness 'includes till 2 synced 06:10' → drill to daily sales by shop → A4 print in Sinhala	A till still offline: the tile shows the stale count and the figures exclude it explicitly
6.3 Federation shrinkage view	Shrinkage % by shop for the month from loss_fact and negative-lot clearances over sales; freshness per MPCS; drill to loss by category; export to spreadsheet	– (view-only by construction)
6.4 Late sync after a two-day outage	Receipt events for two closed business dates arrive → facts inserted; sales_daily for those dates updated; freshness advances; dashboards that showed 'not yet synced' now include the shop; previously rendered reports keep their stamp (a re-run gets the new figures)	–
6.5 A wrong projection	A bug in the loss consumer double-counted a category → fix deployed → RebuildProjection(loss_fact, from_seq = 0) replays the archive into a shadow table → swap → reports correct; no figure was ever edited	Rebuild longer than the window: projection marked REBUILDING; reports show 'temporarily unavailable' rather than wrong figures
6.6 Analyst query	Named analyst connects to the replica with app_analyst; session start audited; reads reporting.* under FEDERATION_VIEW; a query on an operational schema fails on grant	–
6.7 Exception review	MPCS accountant opens the exception queue: negative lots, offline limit breaches, variance beyond threshold, gap anomalies; acknowledges with dispositions (kernel command); unacknowledged ALERTs escalate after 3 days	–

7. Configuration

Item	Default	Scope	Who may change
reporting.freshness_stale_threshold_hours	24	Federation	rpt.definition.manage
reporting.projection_lag_alert_seconds	300	Federation	DevOps
reporting.export_row_cap / pages	500,000 rows / 100,000 per sheet	Federation	DevOps
reporting.export_retention_days	7	Federation	DevOps
reporting.end_of_day_window_minutes (before ALERT)	120	Federation	DevOps (doc 19 DR-3 trigger input)
reporting.dashboard_tiles (per role template)	as seeded	Federation	rpt.definition.manage
reporting.monthly_rollup_time	02:30 Asia/Colombo	Federation	DevOps

8. Screens as paper

Screen	Paper mirrored	Surface	Roles	Notes
Dashboard	Day sheet	W	By scope	Fixed tiles with FreshnessStamp; drill to reports
Catalogue	Standard forms index	W	By audience	Grouped by decision; parameters form; run history
Report viewer	The printed form	W	By audience	A4 render, data view, export; freshness banner
Exception queue	Exceptions book	W	Named reviewers	Shared component; acknowledgement dispositions
Freshness and fleet	–	W	Federation, DevOps	Which shops synced when; stuck devices; projection lag
Definition editor	–	W	Federation	Decision, audience, scope, parameters, source, templates; fixture preview
Analyst access	–	W	Federation admin	Issue and revoke app_analyst credentials; session audit

9. Non-functional design

Aspect	Design
9.1 Latency	Projection lag under 60 s in steady state at 1,000 events/s; during reconnection bursts projections lag and the stamp says so; dashboards read the replica (replica lag added to freshness).
9.2 Volume	Receipt facts: about 3 million rows a day at the horizon, lines 15 million; partitioned monthly; roll-ups keep dashboards on small tables; indefinite retention with cold partitions detached to object storage and re-attachable for historical runs.
9.3 Columnar trigger	When end-of-day aggregation for the federation exceeds <code>reporting.end_of_day_window_minutes</code> on the replica, or dashboard queries exceed 2 s at p95, a columnar store is introduced (doc 19 DR-3); because facts are rebuildable from the archive, it is a rebuild, not a migration.
9.4 Isolation	Consumers run in the worker profile; rendering and exports in the rendering queue; the replica serves reads; the primary bears only projection writes.
9.5 Security	RLS on every projection; the analyst role cannot reach operational schemas; every report run and export is audited as a read; exports carry the scope and stamp in the file.
9.6 Privacy	Customer projections carry account ids and balances, never names or phones (names resolve at render time under M7's RLS for the OWN MPCS only).

10. Test strategy

Layer	What is tested
Unit	Every projection consumer against event fixtures from docs 21A–27A; alias resolution; freshness arithmetic; end-of-day totals; ageing and shrinkage formulas
Rebuild equivalence	For random event sequences: live projection equals a from-zero rebuild, row for row
Idempotency	Replaying any event twice changes nothing
Contract	Every command and query; report data equals rendered data; exports respect the caller's scope
RLS matrix	Shop, MPCS, distributor (PARTY masked), Federation view, analyst role; cost columns invisible to PARTY
Scenario	Flows 6.1–6.7; late sync updates closed days and stamps; a wrong consumer fixed by rebuild
Performance	1,000 events/s projection throughput; end-of-day for 10,000 locations under the window; dashboard p95 under 2 s on the replica
Rendering	Each phase-1 definition renders on fixture data in three languages; export layouts paginate correctly

11. Decision requests

#	Point	Proposed	Decider
DR-1	Report catalogue workshop date	Stage 2 week 3–4; definitions land as data without pipeline change	Federation
DR-2	Phase-1 placeholder set	The doc 07 §5 list with decisions and audiences as in §3.4	Federation admins
DR-3	Analyst role issuance	Named Federation analysts only; audited sessions; revocable	Federation, DevOps
DR-4	Freshness stale threshold	24 hours	Federation
DR-5	Export cap and retention	500,000 rows, 7 days	DevOps
DR-6	Reporting tables (CR-28-1 to doc 18: schema reporting as rebuildable, excluded from part A–E invariants)	Adopt with the 'no source of truth' note	Architecture review

12. Traceability

Section	Identifiers
3.1–3.3	FR-REP-010, 020; O-03, O-07; ADR-30; B-10; doc 19 §6, §9
3.4–3.5	FR-REP-040, 050; O-01, O-04, O-05; doc 07 §5; doc 19 §5.3
3.6–3.8	M-04, M-05, M-06; N-06; doc 19 §4.3; doc 20 Appendix A
9	NFR-PRF; doc 19 DR-3; J-06

End of document. Draft for design review; the contract freezes when M9 acknowledges it, targeted end of Stage 2 week 6.
Implementation guide: 28A.